

Trades-Runtime

Universal Analytics & Human Review Layer - Addendum v0.7

Purpose. Make every authorized Trades-Runtime metric, event, receipt, model output and operational relationship queryable and reviewable by humans through reports, graphs, scales, scorecards, probability analyses, forecasts, simulations and performance models.

Core Requirement

If Trades has the authorized data, a human should be able to interrogate it, compare it, model it, visualize it and generate a reviewable report from it.

1. Universal Query Layer

Authorized users should be able to ask natural-language or structured questions across company, region, branch, trade, department, Van, technician era, customer/property/equipment, job, part, time period, operating mode, model version and receipt type. Reports should not require engineering work for every new question.

Example: "Show Plumbing -> HVAC turnovers for Branches 4-8 over the last 90 days, conversion %, realized contribution, receiving labor hours and originating Van, compared with the prior 90 days." Trades resolves the requested dimensions, applies permissions, produces the analysis and preserves the exact query/report definition.

2. Supported Human-Review Outputs

Output family	Examples
Reports	Operational, branch, department, Van, technician-era, executive, exception and audit reports.
Graphs	Line, bar, scatter, distribution, trend, variance and comparison visualizations.
Heat maps / matrices	Cross-department, Chain D coordination, branch, inventory, skill, turnover and failure relationships.
Scales	0-100, 0-1000 or policy-defined normalized scores with visible calculation methodology.
Scorecards	KPI actuals, targets, attainment %, trends, sample size, confidence and context adjustment.
Probability reports	Outcome likelihood, uncertainty range, calibration, evidence strength and alternative scenarios.
Forecasts	Demand, call type, staffing, inventory, labor, revenue, contribution, weather response and capacity.
Models	Labor x cost x profit, dispatch, trajectory, pricebook, inventory, cross-trade, causal and digital-twin models.
Performance reports	Actual vs predicted, before vs after, branch vs branch, Van vs cohort, policy/model version comparisons.
Solution reports	Root causes, ranked interventions, predicted impact, implementation cost and measured results.

3. Four Analytical Levels

Level	Question answered
Descriptive	What happened?
Diagnostic	Why did it happen?
Predictive	What is likely to happen next, with what probability/uncertainty?
Prescriptive	What actions should humans consider, what are the tradeoffs, and what is predicted to happen under each?

4. Any-Metric Composition

The analytics layer should support arbitrary combinations of authorized metrics rather than a fixed dashboard catalog. Examples include labor hours vs realized contribution; technician difficulty vs callback rate; weather severity vs call-type mix; dispatch distance vs first-trip completion; cross-department turnovers vs equipment age; Chain D failures vs customer delay; pricebook change vs margin; or Field Advisor use vs diagnostic outcome.

Metrics may be grouped, filtered, ranked, normalized, trended, correlated, segmented and compared across supported time windows, while preserving sample-size and uncertainty warnings.

5. Probability & Uncertainty Reports

Probability outputs must never be presented as certainty. Reports should distinguish prediction confidence, evidence strength, source quality, cross-source agreement, verification status, sample size and calibration history.

Example: 78% predicted probability of a next-day no-cool surge; evidence strength HIGH; weather-source agreement HIGH; branch-specific historical sample 41 comparable days; forecast status ESTIMATED, not realized.

6. Actual vs Predicted / Performed Reports

Every predictive subsystem should support settlement reports comparing forecast to actual performance. Examples: predicted vs actual call demand, repair duration, parts usage, dispatch outcome, labor cost, margin, cross-trade conversion, customer displacement, callback rate and solution impact.

Performance reporting should distinguish model accuracy from operational outcome. A good operational result does not automatically prove the forecast was accurate, and an accurate forecast does not prove management chose the best response.

7. Universal Scope & Time Controls

Dimension	Examples
Organization	Company, region, branch, department, trade, team.
People/operations	Van, technician era, manager role, dispatch group, warehouse group.
Assets	Customer, property, equipment, component, part/SKU, inventory location.
Workflow	Booking, dispatch, field, turnover, warehouse, warranty, callback, Chain D handoff.
Mode	Watcher, Shadow, Recommendation, Human-Approved, Limited Autonomy, Autonomous.
Time	Hour/day/week/month/quarter/year; frozen KPI windows 1 week, 1, 3, 6, 9 and 12 months.
Version	Model, policy, pricebook, inventory snapshot, ruleset, deployment and branch maturity state.

8. Drill-Down to Evidence & Receipts

Every chart, score, probability, report and model should permit authorized drill-down from aggregate result to the underlying calls, events, receipts, evidence packets and calculation lineage. Human reviewers should be able to determine where the number came from rather than treating the visualization as an opaque truth source.

Aggregate -> cohort -> event/job -> decision receipt -> evidence packet -> source/version.

9. Mandatory Report Metadata

Field	Required disclosure
Scope / filters	Exactly what records, branches, roles, modes and dates were included/excluded.
Data sources	ServiceTitan/ProBooks/Trades-native/connector/manual or other authorized sources.
Sample size	Number of jobs/events/observations supporting the result.
Formula / methodology	How the metric, scale, savings estimate or probability was calculated.
Assumptions	Operational/model assumptions that materially affect interpretation.
Confidence / uncertainty	Model confidence, evidence strength and relevant interval/range.
Status	Observed, estimated, attributed, reconciled/realized, simulated or unresolved.
Version lineage	Model, ruleset, policy, pricebook and report-definition version where applicable.
Receipt references	Traceable lineage to underlying decision/evidence records.

10. Labor x Cost x Profit Analytics

The analytics layer must treat labor and realized economics as first-class dimensions. Users should be able to graph or model revenue, loaded labor cost, parts, procurement, warranty recovery, callbacks/rework, contribution, margin and contribution per labor hour by any authorized operational dimension.

Profit metrics remain quality- and context-adjusted. Reports must not convert raw profit into an employee-worth score or reward rushing, cherry-picking or inappropriate selling.

11. Cross-Department & Chain D Analytics

The same engine should report HVAC <-> Plumbing <-> Electrical relationship strength, turnovers, validation, conversion and realized contribution, while also analyzing Office/Dispatch/Field/Management/Warehouse/Sales/Accounting coordination health. Users can compare cross-trade opportunity creation with cross-functional communication failure without

mixing the two concepts.

12. Digital Twin & Scenario Reporting

Simulation outputs should be reportable with explicit assumptions and uncertainty. Humans may ask what happens if staffing changes, a branch borrows Vans, prices move, stock is repositioned, maintenance capacity is protected, weather becomes extreme, or a Chain D workflow fix is implemented.

Simulation results must be labeled **SIMULATED/ESTIMATED** until reconciled against real-world results.

13. Report Generation Modes

Mode	Behavior
On demand	Human asks a question and Trades generates the analysis immediately from authorized data.
Saved report	Human saves a reusable query/report definition for later reruns.
Scheduled	Approved reports can run on a defined cadence for management review.
Event-triggered	A report can be generated when an approved condition occurs, such as a major variance or failure threshold.
Settlement	Automatically compare prior prediction/recommendation with realized outcome after the result becomes known.
Audit	Reconstruct a historical decision or metric using point-in-time data and version lineage.

14. Human Review & Decision Boundary

Analytics informs humans; it does not convert probability into fact or automatically authorize high-consequence action. Human reviewers can inspect assumptions, challenge classifications, modify filters, request alternate models, accept/reject proposed solutions and trace every material result back to evidence.

15. Anti-Misleading Visualization Rules

Trades should flag tiny samples, missing data, stale sources, selection bias, incomparable cohorts, estimated values, wide uncertainty and model drift. Scales must expose their definition. Rankings must show sample size/context. Estimated savings must never be displayed as realized savings. Correlation must not be labeled causation without appropriate evaluation.

16. Architecture Integration

Canonical State + Chains A/B/C/D + Cross-Department Matrix + Economics + Weather/Demand + Inventory/Pricebook + Evidence/Receipts + Digital Twin -> Universal Analytics Layer -> Human Review.

The analytics layer therefore becomes the common human interrogation surface across the entire Trades-Runtime architecture rather than another isolated dashboard subsystem.

17. Example Human Queries

Question	Expected result
"Why did Branch 7 margin fall last month?"	Diagnostic decomposition across call mix, labor, parts, callbacks, pricing, staffing and other support
"Graph Van 214 profit quality for 12 months."	Time-series of contribution, labor, margin and quality/context metrics with drill-down.
"Which Chain D failures cost us the most?"	Ranked failure categories with estimated/realized loss and solution status.
"How accurate was Trades dispatch this quarter?"	Shadow/recommendation/autonomous predicted-vs-actual report with model/version splits.

Question	Expected result
"What is the probability Branch 3 needs extra no-coal capacity to improve?"	Before/after causal/probability report with weather/demand evidence and uncertainty.
"Did the warehouse acknowledgment fix work?"	Before/after causal/performance report with labor/cost/profit and recurrence effects.

18. Frozen Requirement

Any authorized metric in Trades-Runtime must be capable of human-readable reporting, visualization, comparison, modeling and evidence-backed review at the appropriate scope. No important model output should exist only as an internal machine value that management cannot inspect and challenge.

End of Trades-Runtime Universal Analytics & Human Review Layer v0.7